

LINUX PROGRAMMING: ASSIGNMENT-2

1. What does the command pwd, whoami , and hostname display?

❖ pwd:

The pwd command prints the full path of your current working directory.

Syntax:

pwd [options]

Options:

-L → prints shortcuts if available.

-P → prints exact canonical path.

bash

pwd

❖ whoami:

The whoami command displays the user-name.

Syntax:

Whoami

It shows the login user name.

bash

whoami

❖ hostname:

To display and even set the system's hostname.

Syntax:

bash

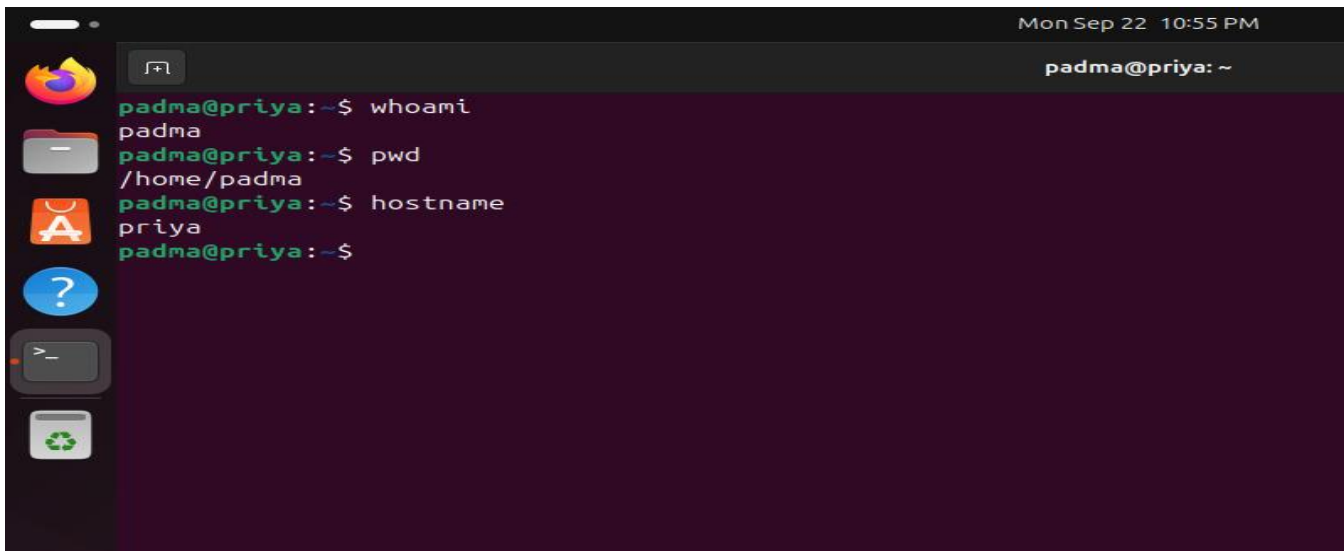
hostname

It directly displays the name.

bash

sudo hostname newuser

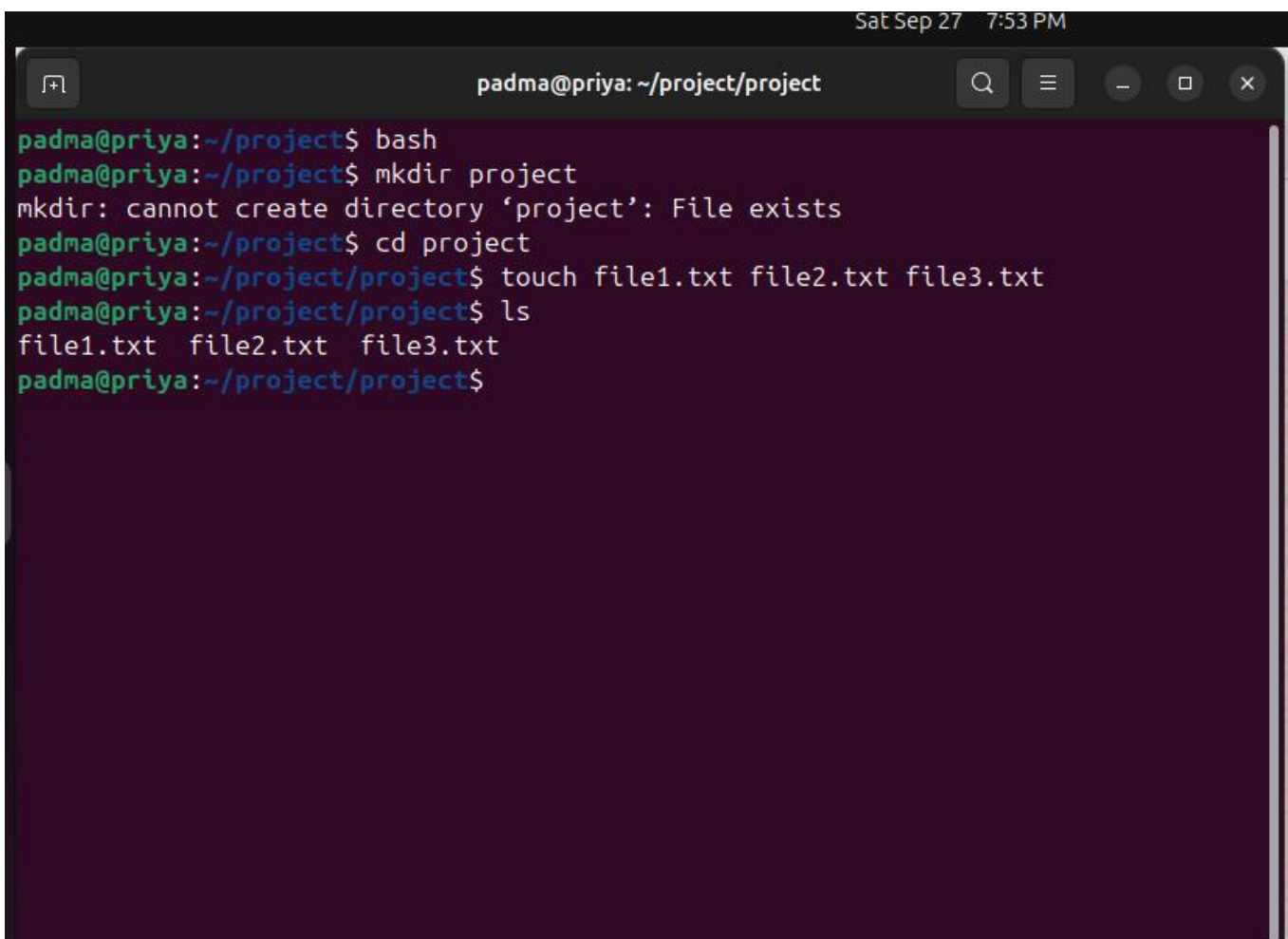
To set the new hostname.



A terminal window titled 'padma@priya: ~' with a timestamp of 'Mon Sep 22 10:55 PM'. The window shows a series of commands and their outputs: 'whoami' returns 'padma', 'pwd' returns '/home/padma', and 'hostname' returns 'priya'. The terminal has a dark purple background and a sidebar on the left with various application icons.

```
padma@priya:~$ whoami
padma
padma@priya:~$ pwd
/home/padma
padma@priya:~$ hostname
priya
padma@priya:~$
```

2. Write the command to create a directory named “project” inside the /home/student folder and keep three .txt file into it. Give output snapshot.



A terminal window titled 'padma@priya: ~/project/project' with a timestamp of 'Sat Sep 27 7:53 PM'. The window shows the following commands and outputs: 'bash' changes the prompt to '~/project\$', 'mkdir project' results in an error 'mkdir: cannot create directory 'project': File exists', 'cd project' changes the prompt to '~/project/project\$', 'touch file1.txt file2.txt file3.txt' creates the files, and 'ls' lists them. The terminal has a dark purple background and standard window controls at the top.

```
padma@priya:~/project$ bash
padma@priya:~/project$ mkdir project
mkdir: cannot create directory 'project': File exists
padma@priya:~/project$ cd project
padma@priya:~/project/project$ touch file1.txt file2.txt file3.txt
padma@priya:~/project/project$ ls
file1.txt file2.txt file3.txt
padma@priya:~/project/project$
```

3. Explain the difference between absolute path and relative path with proper examples.

Features	Absolute Path	Relative Path
Start with	/ in Linux, Drive letter in Windows.	Does not start with / or drive name.
Reference point	Root directory.	Current working directory.
Consistency	Always points to the same location.	Depends on where you are currently.
Example	/home/student/project/file.txt	project/file.txt
	Full address of a house.	Directions from where you are right now.

4. What command will give you the already executed command traces in the terminal. Give output snapshot.

bash

history

```
Sat Sep 27 7:28 PM
padma@priya: ~
padma@priya:~$ bash
padma@priya:~$ history
 1  pwd
 2  tree
 3  sudo apt install tree
 4  tree
 5  sudo apt install gedit
 6  sudo apt install gcc
 7  cat array_demo.sh
 8  gedit array_demo.sh
 9  bash array_demo.sh
10  cat array_demo.sh
11  gedit array_demo.sh
12  bash array_demo.sh
13  gedit array_demo.sh
14  bash
15  passwd
16  mkdir padma.c
17  cd padma.c
```

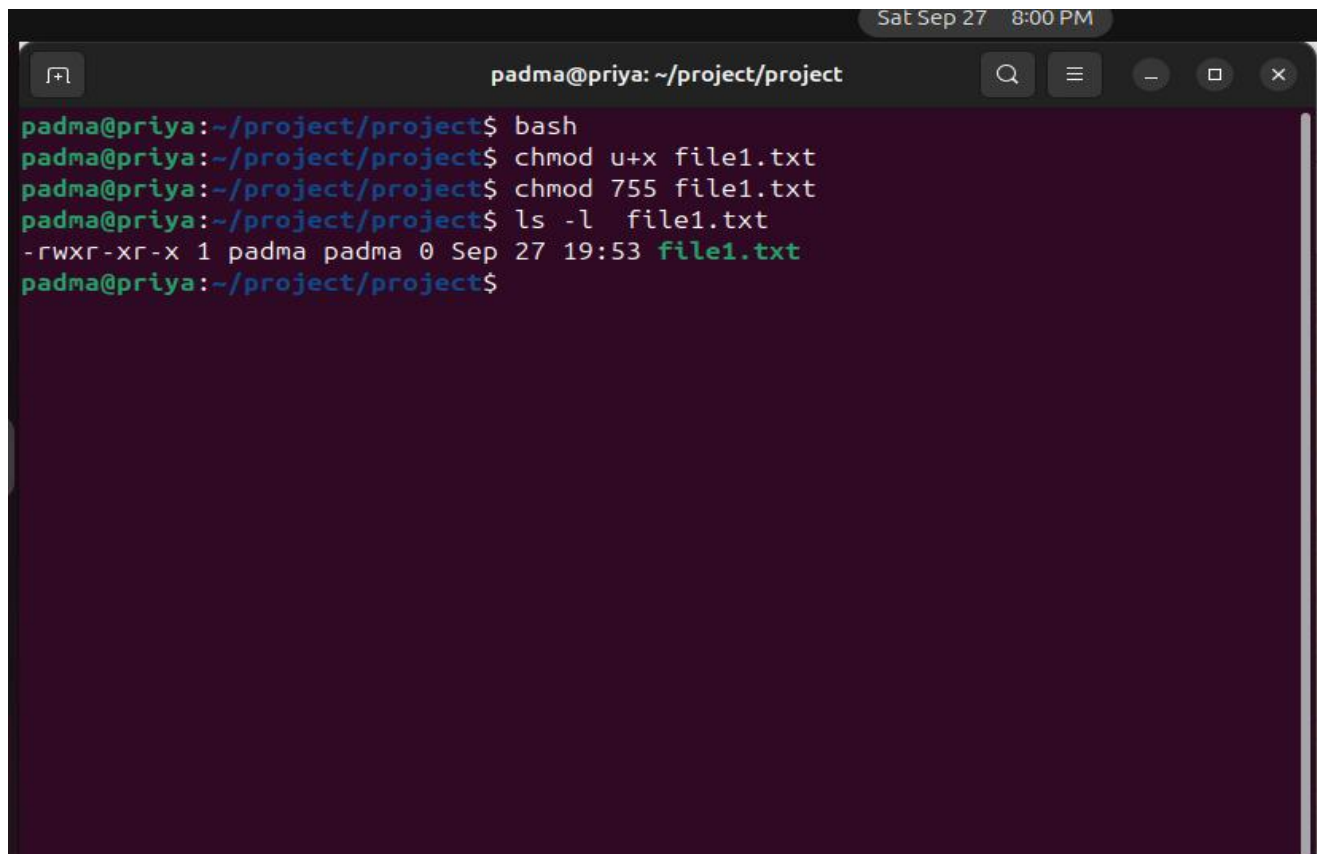
5. Compare the working functionality of **find** and **locate** command. Which one is faster and why?

Features	Find	Locate
Search method	Real-time scanning of directories	Database lookup
Speed	Slower	Faster
Accuracy	Always upto date	Can be out dated
Requirements	No need of any setup	Need mlocate package
Example	find /home -name "file1.txt"	Locate file1.txt

Locate can be faster because it only queries prebuilt database rather than scanning the file system completely, so due to this function locate can be faster compared to find.

6. Which command is used to modify file permissions in Linux? Give an example.

❖ In linux chmod command is used to modify file permission .

A terminal window titled 'padma@priya: ~/project/project' with a search bar and window controls. The terminal shows a sequence of commands: 'bash', 'chmod u+x file1.txt', 'chmod 755 file1.txt', and 'ls -l file1.txt'. The output of the last command is '-rwxr-xr-x 1 padma padma 0 Sep 27 19:53 file1.txt'.

```
padma@priya:~/project/project$ bash
padma@priya:~/project/project$ chmod u+x file1.txt
padma@priya:~/project/project$ chmod 755 file1.txt
padma@priya:~/project/project$ ls -l file1.txt
-rwxr-xr-x 1 padma padma 0 Sep 27 19:53 file1.txt
padma@priya:~/project/project$
```

7. A file has permissions **-rw -r- -r- -**. What does this mean?

-rw -r- -r- -

- ❖ - : normal file
 - ❖ -rw: user permissions
 - ❖ -r-: owner can only read
- Members can't read ,write or excute

8. Explain the difference between chown and chgrp with an example.

Features	chown	Chgrp
Purpose	Changes the owner of file/directory	Changes only the group members of file/directory
Effect of owner	Owner can be changed	Owner can't be changed
Effect on Group	Groups are optional	Groups only change

```
Sat Sep 27 8:20 PM
padma@priya: ~/project/project
padma@priya:~/project/project$ bash
padma@priya:~/project/project$ ls -l file2.txt
-rw-rw-r-- 1 padma padma 0 Sep 27 19:53 file2.txt
padma@priya:~/project/project$ sudo chown padma file2.txt
padma@priya:~/project/project$ ls -l file2.txt
-rw-rw-r-- 1 padma padma 0 Sep 27 19:53 file2.txt
padma@priya:~/project/project$ sudo chgrp padma2 file2.txt
chgrp: invalid group: 'padma2'
padma@priya:~/project/project$ ^C
padma@priya:~/project/project$ sudo chgrp padma file2.txt
padma@priya:~/project/project$ ls -l file2.txt
-rw-rw-r-- 1 padma padma 0 Sep 27 19:53 file2.txt
padma@priya:~/project/project$
```

9. A **file** needs to be accessible by **multiple users** but only **writable** by the owner.

How will you set **permissions**?

- ❖ Owner :read + write(rw-)
- ❖ Group: read only(r--)
- ❖ Others: read only(r--)

-rw-r- -r- -

10. How do you **check the manual page** for any Linux commands?

- ❖ To check the manual page we have to use man command in linux

bash

man ls

man tree

man chmod

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